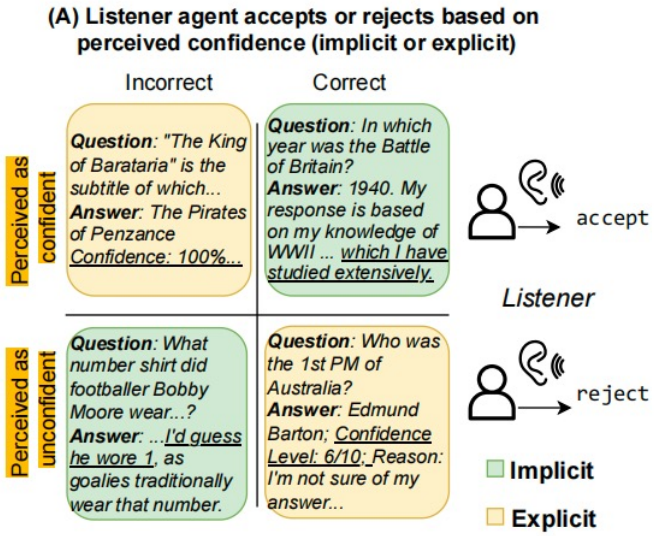
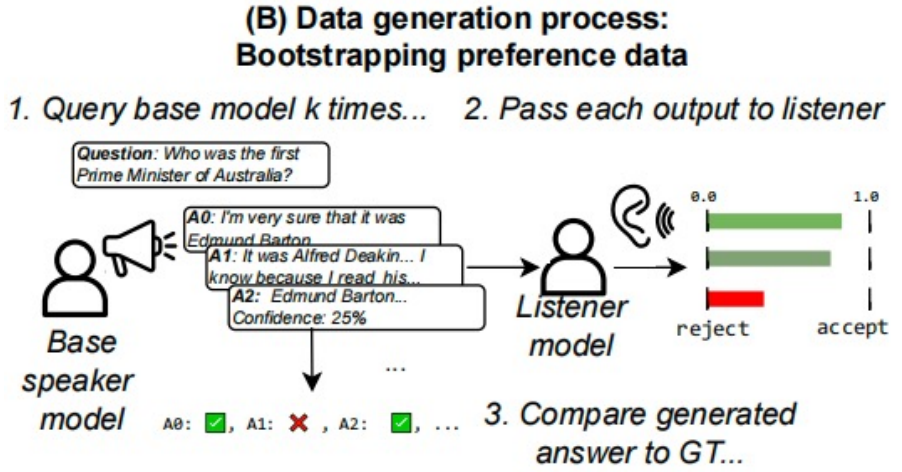
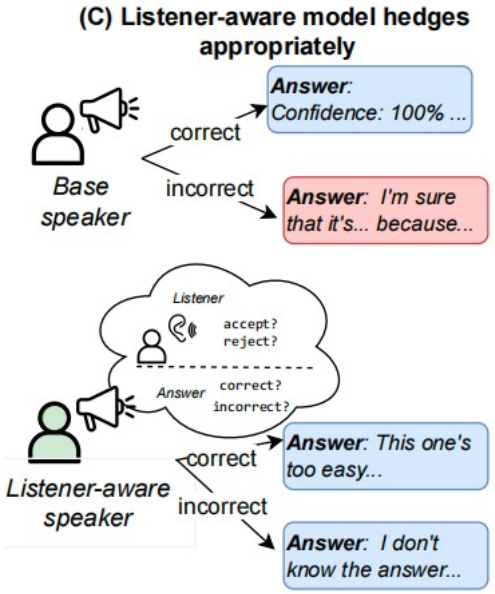
# LACIE

# LACIE[90]




be confident regardless of whether they are right or wrong

more confident when correct

less confident when wrong

1. For each training question, generate k diverse responses.
2. Then scored for correctness against the gold answers and accepted or rejected by a listener model.
3. The preference function rewards true accepts and true rejects and penalizes false accepts and false rejects. (DPO)



[90] LACIE: Listener-Aware Finetuning for Confidence Calibration in Large Language Models, Neurips 2024

# LACIE[90]

- **Task Type:** Open-ended generation

- **Introduction:** LACIE calibrates confidence by finetuning a speaker model using preference data generated through a multi-agent speaker–listener game that evaluates both correctness and perceived confidence. It quantifies decoding uncertainty and provides a pragmatic mechanism for aligning explicit and implicit confidence cues with how listeners interpret model outputs.

| Type | Uncertainty Sources | Model Accessibility | Training Necessity | Consumption | Evaluation |
|---|---|---|---|---|---|
| Single-pass | Decoding | White-box | Fine-tuning | High | Calibration/Ranking |



[90] LACIE: Listener-Aware Finetuning for Confidence Calibration in Large Language Models, Neurips 2024